MILITARY SPECIFICATION

SWIitches, SENSITIVE

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 scope. - **This** specification describes the requirements for the procurement of three types of sensitive switch for use in shipboard fire control equipment. The switches covered by this specification are mechanically actuated, snap-action type, with small travel, for use in either alternating-current or direct-current circuits. Type-I switches are single-pole, double throw, with a roller plunger actuator. Type-II switches are dual single-pole, double-throw, switch assemblies, with a common leaf spring actuator. Type-111 switches are single-pole, double-throw, with a roller leaf spring actuator.

1.2 Classification. - The sensitive switches shall be of the following types:

Type	BUWEPS Drawing No. **1981293**	Switch, Sensitive
Type II	BUWEPS Drawing NO. **2031831**	Actuator, Switch
Type III	BUWEPS Drawing No. **1885125**	Switch, Sensitive with Roller Adapter

2. APPLICABLE DOCUMENTS

291. The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation of bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

FSC 1290

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Boxes, Folding, Paperboard

Boxes, Wood, Cleated-Plywood

Boxes, Wood, Nailed and Lock-corner

Boxes, Fiber

Boxes, Setup, Paperboard

Preservation, Methods of

Shockproof Equipment Class HI (High Impact) Shipboard Application, Tests for

Boxes, Paperboard, Metal Stayed

Environmental Testing, Aeronautical and Associated Equipment, General Specification for

Fire Control, Optical and Associated Equipment Intended for Shipboard Use, the General Specification for Design and Procurement of

Weapons, Handling and Preparation for Delivery of; General Requirements for

Marking for Shipment and Storage Equipment

Mechanical Vibrations for Shipboard Equipment

Switch, Sensitive

Actuator, Switch

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BUWERS Dwg. No. 1885125 — Switch, Sensitive, with Roller Adapter

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVORD OS 6341 — Miscellaneous General Ordnance Design Requirements

(COpies of specifications, standards, drawings, and publications requested by contractors in connection with specific procureMENt functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF' STANdaRDs PUBLICATIONS

Handbook H28 — Screw-thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION lNC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W., Washington 6, D.C.)

30 REQUIREMENTS

3.1 General requirements.

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3.1.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the sensitive switch shall be manufactured using the methods and procedures proposed for the production lot . The sample will be tested as specified in section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding Items that comply with the technical requirements of the contract. Provision of samples shall be in accordance with 4.1.1.

3.1.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the sensitive switch shall conform to the detailed requirements given on the applicable BUWEPS drawing and In the specifications listed thereon, and in Specification MIL-F-18870. Material not covered by this specification, or not specifically designated herein, shall be of the best quality, of the lightest weight consistent with strength and durability, and entirely suited to the purposes for which they are intended under all conditions of test and service operation.

Metals shall be of a corrosion-resisting type or shall be suitably plated to resist corrosion. The use of dissimilar metals in contact which tend toward active electrolytic corrosion (particularly brass, copper, or steel in contact with aluminum or aluminum alloy) is not acceptable. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The use of dissimilar metals separated by a suitable non-metallic, insulating material is also permitted.

3.1.3 Finishes, coatings, construction, and dimensions. - Finishes, coatings, construction, and dimensions for the sensitive switch shall be in accordance with the applicable BUWEPS drawing and with the specifications and publications listed thereon.

3.1.4 Moisture, fungus, and corrosion. - The sensitive switch shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the applicable BUWEPS drawing and in the specifications listed thereon.

3.1.5 Marking. - Marking of assemblies, sub-assemblies, and parts shall be in accordance with the applicable BUWEPS drawing.

3.1.6 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the requirements shown on the applicable drawings . If so stipulated in the contract , order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in design of the contractor's inspection gages. However, such

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procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Naval Final Inspection Gages.

3.1.7 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.1.8 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effeet on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.1.9 Conflicting requirements. - Conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2 Definitions. - Definitions applicable to this specification and associated drawings for single-pole sensitive switches are as follows:

3.2.1 Operate force. - The amount of force, in ounces, which is applied in the line of the plunger or actuator motion to cause a transfer of the switch contacts on the actuating stroke. See figure 1.

3.2.2 Release force. - The amount of force, in ounces, to which the operate force must be reduced to cause a transfer of the switch contacts on the release stroke with or without pretension on the plunger or actuator. See figure 1.

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3.2.3 Free position. - The unoperated, fully extended position of the plunger or actuator. See figure 1.

3.2.4 Operate position. - The position of the plunger or actuator when the transfer of the switch contacts occurs on the actuating stroke. See figure 1.

3.2.5 Overtravel position. - The final position of the plunger or actuator at the end of its actuating stroke after the plunger or actuator is fully depressed. See figure 1.

3.2.6 Release position. - The position of the plunger or actuator when the transfer of the switch contacts occurs on the return stroke. See figure 1.

3.2.7 Pretravel. - The distance traveled in moving the plunger or actuator from the free position to the operate position. This distance is usually expressed as a maximum figure, which means that the distance specified will not be exceeded. See figure 1.

3.2.8 Overtravel. - The distance traveled in moving the plunger or actuator from the operate position to the overtravel position. This distance is usually expressed as a minimum figure. See figure 1.

3.2.9 Movement differential. - The distance traveled by the plunger or actuator in moving from the operate position to the release position on the release stroke. This is affected by the amount of contact separation. This distance is usually expressed as a maximum figure, which means this distance will not be exceeded. See figure 1.

3.3 Performance requirements and product characteristics. - The sensitive switch shall conform to the following performance requirements and product characteristics:

3.3.1 Electrical requirements.

3.3.1.1 Dielectric strength. - There shall be no dielectric breakdown when the sensitive switch is tested with 1000 volts in accordance with ***4.4.4.1.***

3.3.1.2 Insulation resistance. The insulation resistance of the sensitive switch shall not be less than 500 megohms when tested in accordance with 4.4.4.2.

3.3.1.3 Contact resistance. - The contact resistance of the sensitive switch shall not exceed 0.025 ohm when tested in accordance with 4.4.4.3.

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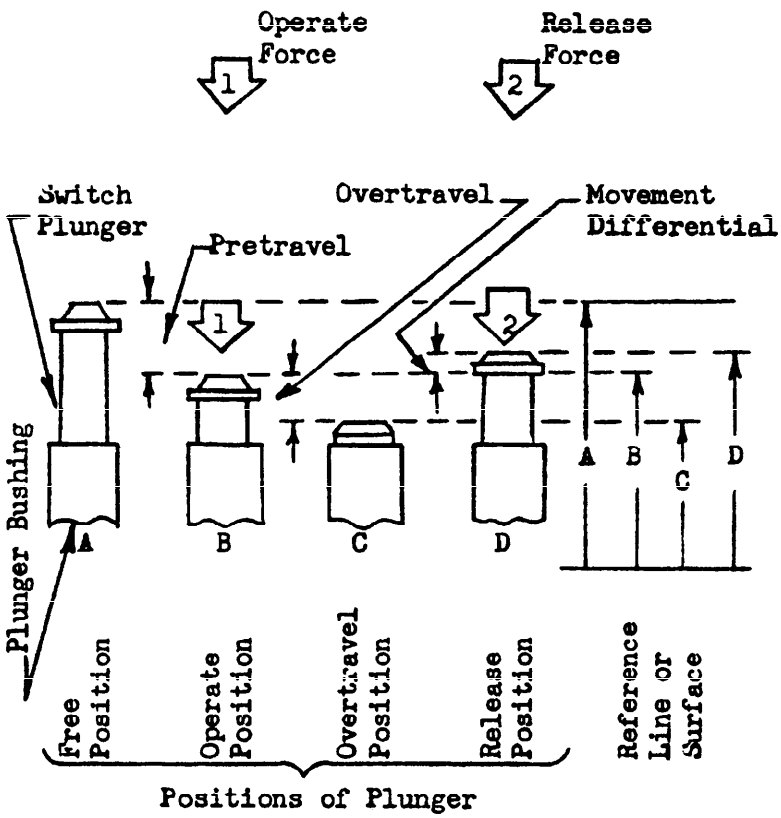


Figure 1. Sequence of Operation

3.3.2 Mechanical requirements.

3.3.2.1 Operate force. - The operate force shall be 10 ounces maximum for type II, when the sensitive switch is tested in accordance with 4.4.4.4.

3.3.2.2 Release force. - The release force shall be 4 ounces maximum for type I, 10 ounces minimum for type II and 1 ounce minimum for type III when the sensitive switch is tested in accordance with 4.4.4.4.

3.3.2.3 Operate position. - The operate position shall be 0.780 plus or minus 0.030 inch for type I, 0.468 plus or minus 0.005 inch for type II and 0.562 plus or minus 0.015 inch for type III when the sensitive switch is tested in accordance with 4.4.4.5.

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3.3.2.4 Pretravel. - The pretravel shall be 0.035 inch maximum for type I, 0.180 inch maximum for type II and 0.0313 inch nominal for type III when the sensitive switch is tested in accordance with 4.4.4.6.

3.3.2.5 Overtravel. - The overtravel shall be 0.035 inch minimum for type I, 0.070 inch minimum for type II and 0.010 inch minimum for type III when the sensitive switch is tested in accordance with 4.4.4.7.

3.3.2.6 Movement differential. - The movement differential shall be 0.005 inch nominal for type I, 0.015 inch maximum for type II and 0.003 inch maximum for type III when the sensitive switch is tested in accordance with 4.4.4.8.

3.4 Environmental requirements. - The sensitive switch shall meet the other applicable requirements of this specification during or after tests defined or' specified in 4.5.

3.4.1 Vibration

3.4.1.1 Vibration, transportation. - The sensitive switch shall be capable of withstanding the vibration tests specified in 4.5.1.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.1.2 Vibration, operating condition. - The sensitive switch shall withstand the type I vibration requirements specified in Standard MXL-STD-167, shall function during and after vibration tests, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2 Shock

3.4.2.1 Shock, transportation, storage and handling. - The sensitive ' switch shall be capable of withstanding the shock tests specified in 4.5.2.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.2.2 Shock, non-operating condition. - The sensitive switch shall withstand the type A shock requirements specified in Specification MIL-S-901, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.3 Temperature.

3.4~3.1 Temperature, operating condition. - The sensitive switch shall withstand operational ambient temperatures of between minus 67 degrees Fahrenheit (minus 55 degrees centigrade) and 257 degrees Fahrenheit **(1.25 degrees**

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centigrade for type I, minus 67 degrees Fahrenheit (minus 55 degrees centigrade and 185 degrees Fahrenheit (85 degrees centigrade) for type II and type III in accordance with Specification MIL-F-18870, shall function during and after temperature tests, and thereafter shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of this specification.

3,4.3.2 Temperature, non-operating condition. - The sensitive switch shall withstand storage temperatures of between minus 85 degrees Fahrenheit (minus 65 de grees centigrade) and 257 degrees Fahrenheit (125 degrees centigrade) in accordance with Condition II of os 6341, and thereafter shall show no corrosion or damage and shall meet all the performance characteristics of this specification.

3.4.4 Humidity. - The sensitive switch shall withstand operational ambient humidities in accordance with Specification MIL-F-18870, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.5 Tropical environment. - The sensitive switch shall withstand the tropical environmental conditions specified in Specification MIL-E-5272 under "Fungus Resistance 'Tests", and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics of this specification.

3.4.6 operating life. - The sensitive switch shall withstand 25,000 cycles of operation when tested as specified in 4.5.6, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5 Clean-up. - The assembly shall be **thoroughly cleaned of loose metal** chips, and other foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship. - The assembly shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of marking of assemblies, machine screw assemblage, and freedom of parts from burrs and sharp edges.

3.6.1 In addition to the examination and test performed on the pre-production samples, the workmanship exhibited therein will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirements for all units offered for acceptance.

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40 QUAlity assurance PROVISIONS

4.1 Unless Otherwise specified in the contract or purchase **order} the supplier is responsible** for the performance of all Inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any commercial laboratory acceptable to the Government. The Government ***reserves*** the right to **perform any of** the inspections set forth in the **specification were such Inspections are deemed necessary to assure** supplies and services conform to prescribed requirments.

4.1.1 Pre-production samples. - Unless otherwise specified for each contract or order, a representative sample of three sensitive switches shall be manufactured by the methods and procedures proposed for the production lot. These shall be submitted for pre-production tests at an activity designated by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the Pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part **of** the quantity specified by the contract or order, **but shall not be used In any equipment manufactured for the Government.**

4.1.2 Periodic production samples. - The tests of 4.4 and k.5 shall be performed in steps as specified below. Periodic product ion samples ac - cepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

step 16 - Upon initiation of' production, select one (1) unit from **each** group of ten (10) **units for submission to an activity** designated by **the procuring agency.**

step **2. -** Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each **group** of twenty-five (25).

step 3. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group **of fifty (50) units .** Continue this procedure until a defective unit is **found .**

step **4• -** When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to step 1 and proceed as indicated.

4.2 Classification of tests. - The inspection and testing of the sensitive switch shall be classified as follows:

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4.2.1 Acceptance tests. - These tests shall be accomplished on each sensitive switch submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production **and periodic production tests. - Pre-production and periodic production tests shall be accomplished on samples selected as specified in 4.1.1 and 4.1.2 which are representative of the production of the sensitive switch after the award of the contract, to determine that the production meets the requirements of this specification. These** tests are detailed in 4.5. Acceptance shall be based on no defects in the sample. **Failure of** the sample to comply with these requirements will result in the rejection of the lot or the cessation of production, as determined by the procuring activity.

4.2.3 Tests **after delivery. - Unless ottherwise specified in the contract, the Governmennt reserves the right to conduct any or all the tests specified herein, as deemed necessary, after Installation of the sensitive** switch. Any **defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications** for this equipment shall be corrected by the contractor at 'no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth In this specification.

4.3.1 High potential tester. - A high potential tester capable of applying 1000 volts root-man-square with waveform close to sinusoidal under all test conditions, at a frequency not less than 25 cycles per second (c.p.s.) and not greater than 100 c.p.s. The high potential. tester shall have an indicator which Indicates whenever dielectric breakdown of the unit under test occurs.

4.3.2 Insulation-resistance **tester. - An insulation-resistance** tester **capable of measuring insulation resistance to 1000 megohms with a direct-current test potential not less than 100 volts and not greater than 500 volts.**

4.3.3 **Contact-resistance equipment.**

4:3.3.1 Power supply. - A direct-current power supply having an **output voltage of 7 plus or minus 1 volts, and capable of supplying up** to **5 amperes.**

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4.3.3.2 Rheostat. - A rheostat having resistance adjustable to 10 ohms, and capible of carrying 2 amperes.

4.3.3.3 Direct-current ammeter. - A direct-current ammeter having a **full scale reading of 1 ampere, and an accuracy of plus or minus 2 percent** of full scale .

4.3.3.4 Direct-current millivoltmeter. - A direct-current millivoltmeter having a full scale reading of approximately 25 millivolts, an accuracy of plug or minus 5 percent of full scale, and an input resistance of at least 100 ohms.

4.3.3.5 **Switch. - A single-pole, single-** or double-throw toggle switch rated for 10 amperes minimum, continuous duty, resistive load.

4.3.4 Elastometer. - A test fixture capable of securing the sensitive switch in a vertical position, and of applying a **downward force to the** actuator of the sensitive switch which is proportional to the mass of the weight hung on elastometer weight arm. The value of the constant of proportionality Is not critical, but shall be within plus or minus 1 percent of its nominal value.

4.3.5 Set of weights. - A set of weights which, when used with the elastometer of 4.3.4, will permit application to the actuator of the sensitive switch **of the release** force, **and the difference of the operate force and the release force, as specified on the applicable BUWEPS** drawing. The accuracy **of the weights shall be plus or minus 0. 5 percent.**

4.3.6 Dial indicator. - A dial **indicator which can be mounted on the elastomer of 4.3.4, and which measures to an** *accuracy* **of 0.0001 inch.**

4.3.7 Height gage. - **A height gage which can be mounted on the elastomer of 4.3.42** and which measures to an accuracy of 0.001 inch.

4.3.8 **COntinuity indicator. - An ohmmeter, or lamp with 6 volt power source. Two contInuity indicators are required.**

4.4 Acceptance tests.

4.4.1 **The contractor shall conduct acceptance tests as** specified in 4.2.1 to assure **that the Sensitive switch is in compliance with the re - quirements** of **3.1, 3.3, 3.5, and** 3.6. Manufacturer's certification of materials specified in the applicable drawings shall be used in lieu **of material tests.**

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4.4.2 Test conditions. - Unless otherwise specified, the sensitive switch shall be subjected to acceptance tests under the following conditions :

4.4.2.1 **Temperature:** Room ambient 65 to 95 degrees Fahrenheit (18 to 35 degrees centigrade).

4.4.2.2 Altitude: Normal ground **atmospheric pressure.**

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure. - The detailed test procedure listed in the following paragraphs shall be followed.

4.4.4.1 Dielectric strength. - High potential voltage shall be applied between all pairs of non-connected points for one minute for both the free and operate **positions of the sensitive switch, using the high-potential** tester of 4.3.1, to determine compliance with the requirements of 3.3.1.1.

4.4.4.2 Insulation resistance. - Immediately following the dielectric strength test of 4.4.4.1, the resistance between each pair of non-connected points shall be measured for both the free and operate positions of the sensitive switch, using the insulation-resistance tester of 4.3.2, to determine compliance with the requirements of 3.3.1.2.

4.4.4.3 Contact resistance. - The contact resistance between each pair of mating contacts of the sensitive switch shall be measured, using the **circuit** of figure **2, to determine compliance with the requirements of 3.3.1.3. With the rheostat adjusted to give an ammeter current of 1 ampere, the contact resistance is equal to the reading in volts of' the millivoltmter. The switch of 4.3.3.5 shall be opened only when a millivoltmeter reading is to be taken in order to protect the millivoltmeter from overload.**

4.4.4.4 **Operate force and release** force.

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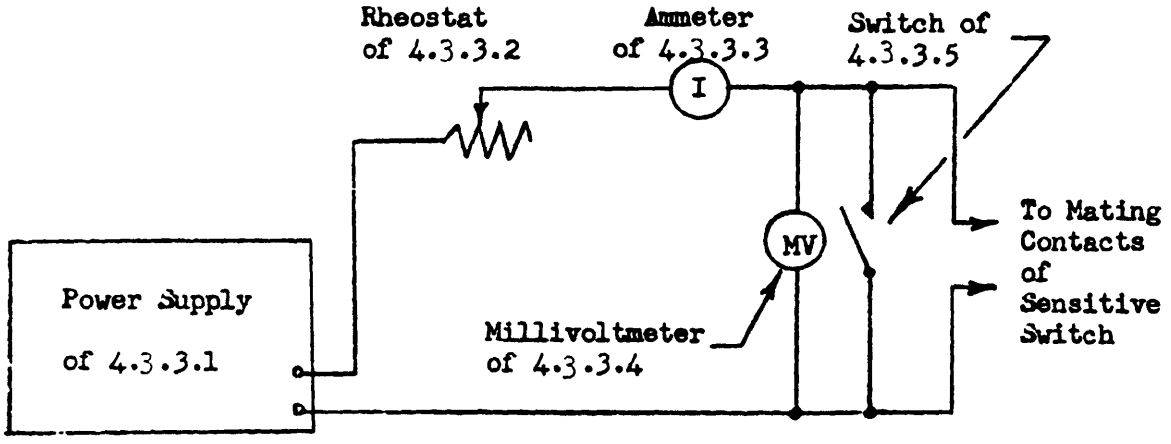


Figure 2. Circuit for Measurement of Contact Resistance

4.4.4.4.1 Sensitive switch types I and III. - With the sensitive switch mounted vertically on the lower platen of the elastometer of 4.3.4, one continuity indicator of 4.3.8 connected between the C and NC switch terminals, and the other continuity indicator connected between the C and NO switch terminals, weights corresponding to the release force and to the difference of the operate force and the release force (see 4.3.5) shall be hung on the elastometer weight arm, and each continuity indicator shall be observed for reversal of indication to determine compliance with the requirements of 3.3.2.1. The weight corresponding to the difference between the operate force and the release force shall then be removed, and each continuity indicator shall be observed for reversal of indications to determine compliance with the requirements for 3.3.2.2.

4.4.4.4.2 Sensitive switch type II. - The procedure of 4.4.4.4.1 shall be followed with the continuity indicators connected to either of the single-pole double-throw switches of the sensitive switch, and then repeated for the continuity indicators connected to the other single-pole, double-throw switch to determine compliance with the requirements of 3.3.2.2.

4.4.4.5 Operate position.

4.4.4.5.1 Sensitive switch type II. - The procedure of 4.4.4.5.2 shall be followed with the continuity indicators connected to either of the single-pole, double-throw switches of the sensitive switch, and then repeated for the continuity indicators connected to the other single-pole, double-throw switch to determine compliance with the requirements of 3.3.2.3.

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4.4.4.5.2 Sensitive switch type 1 and type III. - With the sensitive switch mounted vertically on the lower platen of the elastometer of 4.3.4, the upper platen held fixed, and the continuity Indicators of 4.3.8 connected as specified in 4.4.4.4.1, the lower platen shall be raised slowly until the indication of each continuity indicator reverses. The operate **distance shall then be measured using the height gage of 4.3.7 to determine compliance with the requirements of 3.3.2.3.**

4.4.4.6 Pretravel. - With the sensitive switch mounted vertically on the lower platen of the elastometer of 4.3.4, the upper platen held fixed, the dial gage of 4.3.6 mounted to measure change in the distance between the platens, and the continuity indicators of 4.3.8 connected as specified in 4.4.4.4.1, the lower platen shall be raised slowly until actuator of the sensitive switch makes contact with the upper platens and the reading of the dial indicator corresponding to this separation of the platens shall be recorded. The lower platen shall then be raised slowly until the indication of each continuity indicator reverses, **and the reading of the dial indicator corresponding to this separation of the platens shall be recorded. The difference of the two dial indicator readings shall be noted to determine compliance with the requirements of 3.3.2.4.**

4.4.4.7 Overtravel. - After completing the procedure specified in 4.4.4.6, the lower platen shall be raised slowly until the actuator of the sensitive switch is depressed to the full extent of its travel, and the reading of the dial indicator corresponding to this separation of the platens shall be recorded. The difference between this dial indicator reading and the second dial indicator reading obtained in 4.4.4.6 shall be noted to determine compliance with the requirements of 3.3.2.5.

4.4.4.8 Movement **differential. - After completing the procedure specified in** 4.4.4.7, the **lower platen shall be lowered slowly until the** indication of each continuity indicator reverses, and the reading of the dial indicator corresponding to this separation of the platens shall be recorded. The difference between this dial indicator reading obtained in 4.4.4.6 shall be noted to determine compliance with the requirements of 3.3.2.6.

4.4.5 Presentations packaging and packing. Prese-tion, Packaging and packing of the sensitive switch shall conform to section 5 of this specification.

4.5 Pre-production **and periodic production sample tests.**

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4.5.1 Vibration.

4.5.1.1 Vibration, transportation. - The sensitive switch, packaged and packed as specified in section 5, shall be vibration tested as specified in Specification MIL-W-21927.

4.5.1.2 Vibration, operating condition. - The sensitive switch, rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests so as to meet the requirements specified in 3.4.1.2 and shall meet all the performance requirements and product characteristics of this specification during and after the vibration tests.

4.5.2 Shock.

4.5.2.1 Shock, transportation, storage and handling. - The sensitive switch, packaged and packed as defined in section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

4.5.2.2 Shock, non-operating condition. - The sensitive switch, rigidly mounted in a suitable shock fixture, shall be subjected to shock tests so as to meet the requirements specified in 3.4.2.2 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3 Temperature.

4-5.3.1 Temperature, operating condition. - The sensitive switch shall be subjected to temperature tests so as to meet the requirements specified in 3.4.3.1 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3.2 Temperature, non-operating condition. - The sensitive switch shall be subjected to temperature tests so as to meet the requirements specified in 3.4.3.2 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.4 Humidity. - The sensitive switch shall be subjected to humidity tests so as to meet the requirements specified in 3.4.4, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.5 Tropical environment. - The sensitive switch shall be exposed to fungus tests so as to meet the requirements specified in 3.4.5, and thereafter shall meet all the performance requirements and product characteristics of this specification.

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4.5.6 operating life.

4.5.6.1 Sensitive switch types I and III. - The sensitive switch shall be operated for 25,000 cycles using the circuit of figure 3 so as to meet the requirements specified in 3.4.6, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.6.2 Sensitive switch type II. - The sensitive switch shall be operated for 25,000 cycles using the circuit of figure 4 so as to meet the requirements specified in 3.4.6, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6 Wavier of tests. - Tests **specified in 4.5.1.1 and 4.5.2.1 may be waived** by the **procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.**

4.7 Functional tests. - At the conclusion of the tests specified in 4.5.l.l and 4.5.2.1, functional and other tests, as applicable, shall be conducted on the sensitive switch to determine freedom from operational malfunction and compliance with all the requirements of section 3.

4.8 Failure of the sensitive switch to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATIon FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 level A.

5.1.1.1 Cleaning. **-Not** applicable.

5.1.1.2 Unit packaging. - Unit packaging of the sensitive switch shall be in accordance with Method IA-8 of Specification MIL-P-116. THe unit package shall contain one sensitive switch.

5.1.1.3 Intermediate package. - Ten sensitive switches packaged as described in 5.1.1.2 shall be Intermediate packaged in paper board boxes in accordance with specification PPP-B-566, PPP-B-676, or MIL-B-4229.

5.1.1.4 Cushioning. - The unit package, intermediate package (when used) and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to ensure that the sensitive switch will meet all the requirements of section 3 after being tested in accordance with 4.5.1.1 and 4.5.2.1.

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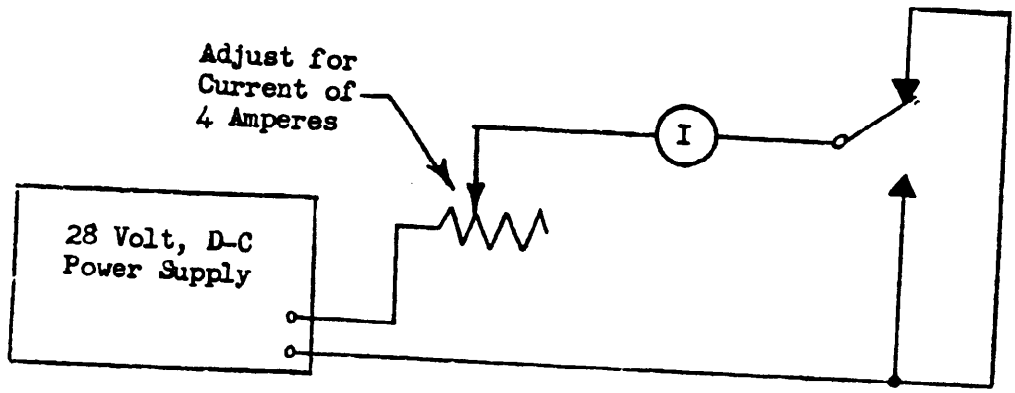


Figure 3. Circuit for Operating Life Test of 4.5.9.1

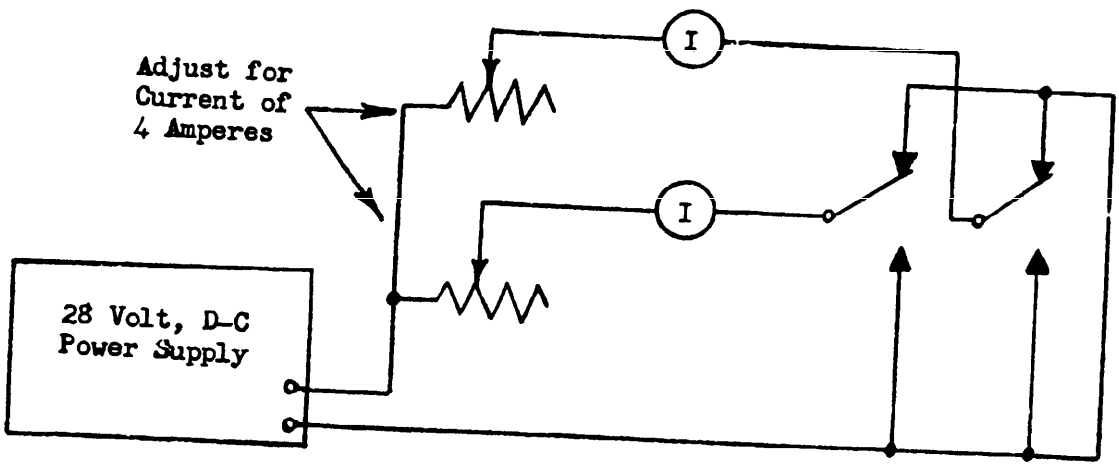


Figure 4. Circuit for Operating Life Test of 4.5.9.2

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

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5.1.3.1 Cleaning, drying and preservative application. - Not applicable.

5.1.3.2 Unit packaging. - Unit packaging of the sensitive switches shall be in accordance with Method 111 of Specification MIL-P-l16-

5.1.3.3 Intermediate package. - Intermediate packages may be used.

5.2 Packing.

5.2.1 Number of packages per pack. - The number of unit packages per **pack shall be as specified in the contract or purchase order.**

5.2.2 Level A.

5.2.2.1 **Exterior containers. - Sensitive switches packaged as described In 5.1.1.2 and 5.1.1.3 shall be packed in fiberboard boxes in accordance with Specification PPP-B-636, Class 2 (weight limit 70 lbs.), wood boxes in accordance with PPP-B-601, overseas type (weight limit 150 lbs) or PPP-B-621, Class 2 (weight limit 150 1bs).**

5.2.3 Level B.

5.2.3.1 **Exterior containers. - Sensitive switches packaged as described in 5.l.1.2 and 5.1.1.3 shall be packed in fiberboard boxes in accordance with Specification PPP-B-636, Class 1 (weight limit 70 lbs), wood boxes in accordance with PPP-B-601, domestic type (weight limit 150 lbs.) or PPP-B-621, Class 1 (weight limit 150 lbs.)**

5.2.4 Level C. - Sensitive switches packaged In accordance with 5.1.3.2 and 5.1.3.3 (when used) shall be packed to afford protection against damage during shipment from the supply source to the first re-ceiving **activity for** immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation-

5.3 Marking.

5.3.1 special markings.- None, unless otherwise specified.

5.3.2 Normal markings. - In addition to the markings requiredby contract or order, unit packages, intermediate packages (when used) and shipping containers shall be marked in accordance with the requirements of MIL-std-129, *Including* **serial numbers.**

6. **NOTES**

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6.1 Intended use. - The sensitive switch is intended for use in the Angle Solver Mark 18 Mod 2 and Mod 3, the Fire Control System Mark 112 Mod 1, the Sonar Receiving Sets AN/BQG-1, -2, -3, -4, or other suitable application.

6.2 Ordering data. - Procurement documents should specify the specific **title,** number, and date **of this specification and exceptions to this specification; applicable drawings and other documents; selections of applicable levels** of preservation, packaging, and packing; and guarantee clause. (See 6.4.)

6.2.1 The attention of the contracting officer is Invited to the Quality Assurance Revisions and options in Specification MIL-P-116 and to para. 4.6.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for six months or less or in domestic locations for an indefinite period.

For level C. - This level shall be used only when it is definitely known that the packaged item Is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. - The contract or order shall **include a** requirement **for a guarantee stating that any sensitive switch which fails to meet the requirements of this specification within one year** after delivery because of a defect in **material or workmanship or because of** the **vendor's failure to** comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor **at** no cost to the Government.

6.5 The contract or order shall include sufficient number of extra items to cover production and pre-production testing samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby Incurs no responsibility nor any obligation whatsoever; and the fact the Government may have formulated, furnished, **or** in any way supplied the said **drawings, specifications, or other data is not to be regarded by Implication or otherwise as In any manner licensing the holder or any other person or corporation,** or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 43-R300

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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